Project Report Day 1

Title: Data Engineering & Ingestion Pipeline

TASK-01: Building a Resilient Pipeline for Live Weather Telemetry

Tools Used :

- VS Code
- Git & Github
- MS Word

Libraries Used

- **requests:** It is used to send a request to the OpenWeather API and fetch weather data.

- **dotenv:** It is used to load my API key from a .env file instead of writing it directly in the code (To keep API key safe).

- **os**: It is used to read the API key stored in the .env file using os.getenv().

- **json:** It is used to store the final processed weather data into a .json file.

- **datetime:** It is used to convert the Unix timestamp received from the API into a human-readable date and time format.

Errors

1. No module named 'requests'

I have installed python 3.13 and 3.10 both versions and the requests was installed on python version 3.13 but script was trying to run using 3.10.

Solution : I ran the script using the correct Python version as “py -3.13 WeatherAPI.py”

2. Incorrect API URL(NameResolutionError)

The API URL was written incorrectly, so the request was going to an invalid address.

Solution: I corrected the URL format to properly include the endpoint, city name, and API key.

3. Wrong city names error

I used incorrect spellings of city names.

Solution: I corrected the city names and used valid city names supported by the API.

4. NameError: json is not defined

I used the json module without importing it.

Solution: I added import json at the top of the code.

Conclusion

This task helped me to understand how to fetch and process real-time data using APIs, handle errors, and store structured data in JSON format.